

**CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY**

Faculty of Technology & Engineering

Devang Patel Institute of Advance Technology and Research (DEPSTAR)

CSUC301 – MACHINE LEARNING

Practical File – ODD Semester 2026-27

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# Practical 1

**Aim:** To learn basic Linux commands for system information, file and directory management, file searching, permissions, backups, and system monitoring in a Linux environment.

## Problem definition:

You have joined **Data Analytics Inc.** as a Linux System Administrator. Your task is to set up the **Marketing Analytics** project environment by creating directories, managing files, monitoring system resources, searching data, and creating backups for the Marketing Analytics team.

## Objectives:

- Identify the current user and working directory.
- Display Linux system information.
- Create and manage directories and files.
- Add and view file contents.
- Search files and text using Linux commands.
- Copy, rename, move, and delete files.
- Create backups and monitor system resources.
- Understand basic Linux file management operations.

## Observation / Output

1. Identify the current user.

```
shalvi@LAPTOP-TF0D01GK:~$ whoami
shalvi
```

2. Find the current working directory.

```
shalvi@LAPTOP-TF0D01GK:~$ pwd
/home/shalvi
```

3. Display system information.

```
shalvi@LAPTOP-TF0D01GK:~$ uname -a
Linux LAPTOP-TF0D01GK 6.18.33.2-microsoft-standard-WSL2 #1 SMP PREEMPT_DYNAMIC Thu Jun 18 21:54:43 UTC 2026
```

4. Check system uptime and date.

```
shalvi@LAPTOP-TF0D01GK:~$ uptime
10:01:21 up 0 min, 1 user, load average: 0.06, 0.01, 0.00
```

```
shalvi@LAPTOP-TF0D01GK:~$ date
Fri Jul 17 10:01:24 UTC 2026
```

5. Create a directory named MarketingAnalytics.

```
shalvi@LAPTOP-TF0D01GK:~$ mkdir MarketingAnalytics
shalvi@LAPTOP-TF0D01GK:~$ ^C
shalvi@LAPTOP-TF0D01GK:~$ cd MarketingAnalytics
```

6. Create subdirectories: RawData, ProcessedData, Reports.

```
shalvi@LAPTOP-TF0D01GK:~$ cd MarketingAnalytics
shalvi@LAPTOP-TF0D01GK:~/MarketingAnalytics$ mkdir RawData ProcessedData Reports
shalvi@LAPTOP-TF0D01GK:~/MarketingAnalytics$ ls
ProcessedData  RawData  Reports
```

7. Create data and report files.

```
shalvi@LAPTOP-TF0D01GK:~/MarketingAnalytics$ touch RawData/customer_data.txt
shalvi@LAPTOP-TF0D01GK:~/MarketingAnalytics$ touch Reports/report1.txt
shalvi@LAPTOP-TF0D01GK:~/MarketingAnalytics$ touch Reports/report2.txt
```

8. Add sample content.

```
shalvi@LAPTOP-TF0D01GK:~/MarketingAnalytics$ cat > RawData/customer_data.txt
CustomerID,Name,City
101,Raj,Ahmedabad
102,Priya,Vadodara
103,Amit,Surat
```

9. Display file contents.

```
shalvi@LAPTOP-TF0D01GK:~/MarketingAnalytics$ echo "Marketing Analysis Report 1" > Reports/report1.txt
shalvi@LAPTOP-TF0D01GK:~/MarketingAnalytics$ echo "Marketing Analysis Report 2" > Reports/report2.txt
shalvi@LAPTOP-TF0D01GK:~/MarketingAnalytics$ cat RawData/customer_data.txt
CustomerID,Name,City
101,Raj,Ahmedabad
102,Priya,Vadodara
103,Amit,Surat
```

10. List all files and directories.

```
shalvi@LAPTOP-TF0D01GK:~/MarketingAnalytics$ ls
ProcessedData  RawData  Reports
shalvi@LAPTOP-TF0D01GK:~/MarketingAnalytics$ ls -l
total 12
drwxr-xr-x 2 shalvi shalvi 4096 Jul 17 10:01 ProcessedData
drwxr-xr-x 2 shalvi shalvi 4096 Jul 17 10:02 RawData
drwxr-xr-x 2 shalvi shalvi 4096 Jul 17 10:02 RawData
drwxr-xr-x 2 shalvi shalvi 4096 Jul 17 10:02 Reports
shalvi@LAPTOP-TF0D01GK:~/MarketingAnalytics$ ls -R
.:
ProcessedData  RawData  Reports
./ProcessedData:
./RawData:
customer_data.txt
./Reports:
report1.txt  report2.txt
```

11. Search for report files.

```
shalvi@LAPTOP-TF0D01GK:~/MarketingAnalytics$ find . -name "*.txt"
./RawData/customer_data.txt
./Reports/report1.txt
./Reports/report2.txt
```

12. Search specific content inside files.

```
shalvi@LAPTOP-TF0D01GK:~/MarketingAnalytics$ grep "Raj" RawData/customer_data.txt
101,Raj,Ahmedabad
shalvi@LAPTOP-TF0D01GK:~/MarketingAnalytics$ grep "Marketing" Reports/report1.txt
Marketing Analysis Report 1
shalvi@LAPTOP-TF0D01GK:~/MarketingAnalytics$ grep -r "Marketing" .
./Reports/report1.txt:Marketing Analysis Report 1
./Reports/report2.txt:Marketing Analysis Report 2
```

13. Copy report files to Backup directory.

```
shalvi@LAPTOP-TF0D01GK:~/MarketingAnalytics$ mkdir Backup
shalvi@LAPTOP-TF0D01GK:~/MarketingAnalytics$ cp Reports/*.txt Backup/
shalvi@LAPTOP-TF0D01GK:~/MarketingAnalytics$ ls Backup
report1.txt  report2.txt
```

## 14. Rename files.

```
shalvi@LAPTOP-TF0D01GK:~/MarketingAnalytics$ mv Reports/report1.txt Reports/FinalReport.txt
shalvi@LAPTOP-TF0D01GK:~/MarketingAnalytics$ ls Reports
FinalReport.txt  report2.txt
```

## 15. Move files between directories.

```
shalvi@LAPTOP-TF0D01GK:~/MarketingAnalytics$ mv Reports/report2.txt ProcessedData/
shalvi@LAPTOP-TF0D01GK:~/MarketingAnalytics$ ls ProcessedData
report2.txt
```

## 16. Delete a file.

```
shalvi@LAPTOP-TF0D01GK:~/MarketingAnalytics$ rmdir EmptyFolder
shalvi@LAPTOP-TF0D01GK:~/MarketingAnalytics$ ls
Backup  ProcessedData  RawData  Reports
```

## 17. Delete an empty directory.

```
shalvi@LAPTOP-TF0D01GK:~/MarketingAnalytics$ mkdir EmptyFolder
shalvi@LAPTOP-TF0D01GK:~/MarketingAnalytics$ rmdir EmptyFolder
shalvi@LAPTOP-TF0D01GK:~/MarketingAnalytics$ ls
Backup  ProcessedData  RawData  Reports
shalvi@LAPTOP-TF0D01GK:~/MarketingAnalytics$ |
```

**Result :**

The Linux project environment was successfully created. Various Linux commands were executed to manage files and directories, search data, perform backup operations, and monitor system resources successfully.

**Key Questions /Analysis:****Q1. What is the difference between absolute and relative paths?**

- Absolute Path: Starts from the root directory (/) and gives the complete location of a file or directory.
- Relative Path: Starts from the current working directory and depends on the current location.

**Q2. Why are Linux permissions important for file security?**

Linux permissions protect files from unauthorized access, control read, write, and execute permissions, and improve system security and data integrity.

**Q3. What is the purpose of the grep command?**

The grep command searches for specific words or patterns inside one or more files and filters command output.

**Q4. How does Linux support efficient file management?**

Linux provides a hierarchical directory structure, powerful file management commands (ls, cp, mv, rm, find, grep), file permissions, and shell scripting for automation.

**Q5. Why are backups important in enterprise environments?**

Backups prevent permanent data loss, help recover files after failures or accidental deletion, protect against malware, and support business continuity.

## Supplementary Problems:

1. Create a shell script to automate project directory creation.

```
shalvi@LAPTOP-TF0D01GK:~$ nano project_setup.sh
shalvi@LAPTOP-TF0D01GK:~$ chmod +x project_setup.sh
shalvi@LAPTOP-TF0D01GK:~$ ./project_setup.sh
Project setup completed successfully.
shalvi@LAPTOP-TF0D01GK:~$ |
```

2. Find the top 10 largest files in the system.

```
shalvi@LAPTOP-TF0D01GK:~$ find ~ -type f -exec du -h {} + | sort -rh | head -10
929M /home/shalvi/hadoop-3.4.1.tar.gz
533M /home/shalvi/hadoop/share/hadoop/tools/lib/bundle-2.24.6.jar
198M /home/shalvi/hadoop/lib/native/libhdfspp.a
79M /home/shalvi/hadoop/share/hadoop/yarn/hadoop-yarn-applications-catalog-webapp-3.4.1.war
74M /home/shalvi/hadoop/lib/native/libhdfspp.so.0.1.0
38M /home/shalvi/hadoop/share/hadoop/client/hadoop-client-minicuster-3.4.1.jar
30M /home/shalvi/hadoop/share/hadoop/client/hadoop-client-runtime-3.4.1.jar
19M /home/shalvi/hadoop/share/hadoop/client/hadoop-client-api-3.4.1.jar
18M /home/shalvi/hadoop/share/hadoop/tools/lib/jdk.tools-1.8.jar
15M /home/shalvi/hadoop/lib/native/libnative-task.a
shalvi@LAPTOP-TF0D01GK:~$ |
```

3. Schedule a backup task using cron.

```
shalvi@LAPTOP-TF0D01GK:~$ crontab -e
no crontab for shalvi - using an empty one
Select an editor. To change later, run select-editor again.
 1. /bin/nano      <---- easiest
 2. /usr/bin/vim.basic
 3. /usr/bin/vim.tiny
 4. /bin/ed

Choose 1-4 [1]: 1
crontab: installing new crontab
```

4. Monitor CPU utilization continuously using top/htop.

```
shalvi@LAPTOP-TF0D01GK:~$ top
top - 15:24:25 up 31 min, 1 user, load average: 0.08, 0.17, 0.08
Tasks: 25 total, 1 running, 24 sleeping, 0 stopped, 0 zombie
%Cpu(s): 0.0 us, 0.0 sy, 0.0 ni, 100.0 id, 0.0 wa, 0.0 hi, 0.0 si, 0.0 st
MiB Mem : 7651.2 total, 6764.0 free, 633.5 used, 448.1 buff/cache
MiB Swap: 2048.0 total, 2048.0 free, 0.0 used, 7017.7 avail Mem
```

| PID  | USER     | PR | NI | VIRT    | RES   | SHR   | S | %CPU | %MEM | TIME+   | COMMAND         |
|------|----------|----|----|---------|-------|-------|---|------|------|---------|-----------------|
| 1480 | shalvi   | 20 | 0  | 8168    | 6000  | 3900  | R | 0.3  | 0.1  | 0:00.01 | top             |
| 1    | root     | 20 | 0  | 24160   | 15412 | 11592 | S | 0.0  | 0.2  | 0:01.03 | systemd         |
| 2    | root     | 20 | 0  | 3180    | 2204  | 2072  | S | 0.0  | 0.0  | 0:00.01 | init-systemd(Ub |
| 7    | root     | 20 | 0  | 3180    | 2044  | 1940  | S | 0.0  | 0.0  | 0:00.00 | init            |
| 50   | root     | 19 | -1 | 50400   | 16796 | 15508 | S | 0.0  | 0.2  | 0:00.28 | systemd-journal |
| 82   | systemd+ | 20 | 0  | 22416   | 14464 | 11968 | S | 0.0  | 0.2  | 0:00.08 | systemd-resolve |
| 91   | root     | 20 | 0  | 35452   | 12352 | 9176  | S | 0.0  | 0.2  | 0:01.03 | systemd-udev    |
| 116  | root     | 20 | 0  | 2888    | 1948  | 1820  | S | 0.0  | 0.0  | 0:00.06 | chronyd-starter |
| 124  | root     | 20 | 0  | 4472    | 3080  | 2832  | S | 0.0  | 0.0  | 0:00.01 | cron            |
| 127  | message+ | 20 | 0  | 8796    | 5356  | 4668  | S | 0.0  | 0.1  | 0:00.08 | dbus-daemon     |
| 171  | root     | 20 | 0  | 44096   | 29920 | 14776 | S | 0.0  | 0.4  | 0:00.34 | networkd-dispat |
| 179  | root     | 20 | 0  | 18440   | 9424  | 8176  | S | 0.0  | 0.1  | 0:00.13 | systemd-logind  |
| 199  | syslog   | 20 | 0  | 220548  | 5948  | 4648  | S | 0.0  | 0.1  | 0:00.13 | rsyslogd        |
| 222  | root     | 20 | 0  | 5492    | 2756  | 2572  | S | 0.0  | 0.0  | 0:00.01 | agetty          |
| 280  | root     | 20 | 0  | 123460  | 32744 | 18136 | S | 0.0  | 0.4  | 0:00.26 | unattended-upgr |
| 283  | _chrony  | 20 | 0  | 20424   | 6432  | 5652  | S | 0.0  | 0.1  | 0:00.10 | chronyd         |
| 305  | _chrony  | 20 | 0  | 12096   | 2424  | 1780  | S | 0.0  | 0.0  | 0:00.00 | chronyd         |
| 422  | root     | 20 | 0  | 3188    | 1112  | 980   | S | 0.0  | 0.0  | 0:00.00 | SessionLeader   |
| 423  | root     | 20 | 0  | 3204    | 1252  | 1108  | S | 0.0  | 0.0  | 0:00.02 | Relay(425)      |
| 425  | shalvi   | 20 | 0  | 6268    | 5532  | 3856  | S | 0.0  | 0.1  | 0:00.13 | bash            |
| 426  | root     | 20 | 0  | 8652    | 5544  | 4696  | S | 0.0  | 0.1  | 0:00.01 | login           |
| 471  | shalvi   | 20 | 0  | 22340   | 13520 | 10900 | S | 0.0  | 0.2  | 0:00.17 | systemd         |
| 473  | shalvi   | 20 | 0  | 22916   | 4120  | 2116  | S | 0.0  | 0.1  | 0:00.00 | (sd-pam)        |
| 501  | shalvi   | 20 | 0  | 6136    | 5416  | 3864  | S | 0.0  | 0.1  | 0:00.03 | bash            |
| 1448 | root     | 20 | 0  | 1857908 | 15428 | 12560 | S | 0.0  | 0.2  | 0:00.04 | wsl-pro-service |

5. Generate a disk usage report for all project directories.

```
shalvi@LAPTOP-TF0D01GK:~$ cd MarketingAnalytics
shalvi@LAPTOP-TF0D01GK:~/MarketingAnalytics$ du -sh *
12K    Backup
8.0K   ProcessedData
12K    RawData
12K    Reports
shalvi@LAPTOP-TF0D01GK:~/MarketingAnalytics$ du -sh * > disk_report.txt
shalvi@LAPTOP-TF0D01GK:~/MarketingAnalytics$ cat disk_report.txt
12K    Backup
8.0K   ProcessedData
12K    RawData
12K    Reports
shalvi@LAPTOP-TF0D01GK:~/MarketingAnalytics$ |
```

### Tools /Technology used:

- 📁 Ubuntu Linux (WSL)
- 📁 Bash Shell
- 📁 Linux Terminal
- 📁 Nano Editor
- 📁 Cron
- 📁 Core Linux Commands (whoami, pwd, ls, mkdir, touch, cat, grep, find, cp, mv, rm, rmdir, du, top)

### Conclusion:

This practical provided hands-on experience with essential Linux commands for system administration. It demonstrated how to create and manage files and directories, search data, perform backup operations, monitor system resources, and automate tasks using shell scripts and cron, building a strong foundation in Linux-based file management and administration.